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The Editors
Digital Discovery
Royal Society of Chemistry

Dear Editors,
16 September 2026

We submit the manuscript entitled “**Target breadth and mutation resilience in African-natural-product-inspired antimalarial chemotypes: a computational analysis**” for consideration as a Full Paper in *Digital Discovery*.

Drug-resistant malaria demands chemotypes that survive the pathogen’s evolution, yet most virtual screening cascades rank compounds against a single static target snapshot. Our contribution is a high-throughput computational triage methodology, not another candidate list. From a scaffold-guided expansion of African natural product and synthetic antimalarial seeds into a 65 856-compound hybrid library (92.6 % whole-molecule novelty with 69.3 % privileged-scaffold retention), we formulate a Resistance-aware Retention Score (RRS) that quantifies per-target affinity retention across wild-type and clinically observed mutant states of four *Plasmodium falciparum* targets (PfDHFR, PfCRT, PfATP4, PfClpP). Multi-parameter optimisation distils the library to 17 mutation-resilient candidates, each validated by an external data-first chain: DEKOIS 2.0 benchmarking, MMV compound enrichment, and sub-2.0 Å redocking RMSD on every target.

This fits your scope on three counts: (i) a high-throughput computational methodology for molecular design, (ii) an advanced data workflow with full provenance, checksums, and machine-readable tables, and (iii) reproducible, reusable outputs for the community.

Data and code: the complete deposit — data, code, raw docking outputs, provenance records, checksum manifest — is public on Zenodo under an MIT licence at <https://doi.org/10.5281/zenodo.22696778>. Editors and referees have immediate access; nothing is embargoed.

The work is original, is not under consideration elsewhere, has been approved by all authors, and there are no conflicts to declare.

We thank you for your consideration.

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